INCH-POUND

MIL-A-27471C (USAF)
30 JULY 1993
SUPERSEDING
MIL-M-27471B (USAF)
13 October 1969

MILITARY SPECIFICATION

ADAPTER, PRESSURE-REDUCER, IN-LINE CRU-43A/A

This specification is approved for use by the department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of oxygen adapter designated CRU-43A/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical

Military

MIL-P-116	Preservation, Methods of
MIL-C-5040	Cord, Fibrous, Nylon
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid And Gas
MIL-M-27274	Mask, oxygen, MBU-5/P
MIL-O-27335	Oxygen System, Survival Container - Oxygen kit, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: The Resources and Logistics Services, SA-ALC/TILDM, Kelly AFB TX 78241-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

STANDARDS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

Federal

FED-STD-595	Colors Used in Government Procurement

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-470	Maintainability Program for Systems and Equipment
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Testing for Engineering Development Qualification, and Production
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
* MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements
MS 27796	Connector - Bayonet, Three Pin, Oxygen Mask
MS 29513	Packing, Preformed, Hydrocarbon Fuel Resistant, "O" Ring

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

Air Force

57B3657	Plate - Mounting of Connector, Oxygen Mask to Regulator, Female Assembly of

(Applications for copies should be addressed to SA-ALC/TILDB, Kelly AFB, TX 78241-5000)

2.2 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society of Mechanical Engineers

ASME B46.1	Surface Texture (Surface Roughness, Waviness, and Lay) (DOD adopted)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. The materials shall conform to the applicable specifications and shall be as specified herein. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality and suitable for the purpose intended. Materials that will come in contact with oxygen shall be entirely free from oil, grease, or any other combustible materials.

3.3.1 Nonferrous materials. Except where ferrous materials are essential, nonferrous materials shall be used for all parts of the adapter.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion caused by fuels, salt spray, or any other atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Elastomers. All elastomers used shall have been manufactured within 12 months of the date the adapter is manufactured. The cure date (quarter and year) shall be stamped on the elastomer component, provided there is sufficient area. The figures shall be at least 3/16 inch high. Cure dates and age limitations shall not apply to silicone. All elastomers shall be free from foreign agents that might cause objectionable odors.

3.3.4 Protective treatment. When materials are used in the construction of the adapter that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design. The adapter shall be designed to adapt high altitude seat kit oxygen system MIL-O-27335 to the MBU-5/P pressure breathing oxygen mask specified in MIL-M-27274. The design of the adapter shall include a pressure reduction mechanism to insure correct mask breathing pressure. The adapter shall conform to figures 1 thru 5. The design shall include a safety feature to provide breathing pressures for all rates of flow at altitudes above 30,000 feet.

3.5 Construction. The adapter shall be constructed so that no parts will become loose in service. It shall be built to withstand the strains, jars, vibrations, and any other conditions incident to shipment, installation and service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

3.5.1 Inlet and outlet dust covers. The inlet and outlet of the adapter shall have covers to prevent the entry of dust and foreign material when not in use. The dust covers shall provide a smooth and positive lock with the parts. The dust covers shall be attached to the adapter body by means of a nylon cord, Type II of MIL-C-5040. The nylon cord shall correspond to color 34127 of FED-STD-595.

3.5.2 Dovetail plate. The adapter shall have a male dovetail plate conforming to figures 1, 3, and 4. The plate shall have a manual lock and a release that are readily accessible for use. The plate shall mate smoothly with AF Drawing 57B3657.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Performance

3.7.1 Pressure reduction characteristics. The adapter shall meet the pressure reduction characteristics specified (see table I).

3.7.2 Overpressure. The adapter shall withstand a pressure of 50 inches of water without leakage except for altitude compensating ports which shall not exceed 100 cubic centimeters per minute as specified (see 4.6.3).

3.7.3 Low temperature operation. While subjected to an inlet pressure of 3.5 $\pm$ 0.05 inches of water, the adapter shall operate satisfactorily at -65° $\pm$ 2° F as specified (see 4.6.4). Upon conclusion of this test and while still at -65° $\pm$ 2° F, the adapter shall meet the requirements as specified (see 3.7.1).

3.7.4 High temperature operation. While subjected to an inlet pressure of 3.5 $\pm$ 0.05 inches of water, the adapter shall operate satisfactorily at 160° $\pm$ 2° F as specified (see 4.6.5). Upon conclusion of this test and while still at 160° $\pm$ 2° F, the adapter shall meet the requirements as specified (see 3.7.1).

3.7.5 Low temperature exposure. After continuous exposure to a temperature of -85° $\pm$ 5° F for 48 hours, the adapter shall pass the tests specified (see 4.6.6).

3.7.6 High temperature exposure. After continuous exposure to a temperature of not less than 160° F for 7 hours, the adapter shall pass the tests specified (see 4.6.7).

3.7.7 Altitude cycling. At 0 flow rate or inlet pressure, the adapter shall be cycled 100 times at an altitude from 0 feet to 50,000 feet and back to 0 feet in not less than 30 minutes. The adapter shall then meet the tests specified (see 4.6.8).

3.7.8 G force. When subjected to forces of 6g, the adapter shall operate as specified herein. After being subjected to a force of 22g for a period of 0.01 second, the adapter shall meet the tests specified (see 4.6.9).

3.7.9 Orientation. The adapter shall operate as specified herein when tested in vertical and horizontal positions (see 4.6.10).

3.7.10 Reliability. The adapter shall have a minimum mean time between failure (MTBF) of 870 hours with a minimum reliability of 0.9816 for a 16 hour mission at a confidence level of 0.90 (see 4.6.11).

3.7.10.1 Longevity. All reparable adapters shall have a minimum operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2700 hours before wear out failures occur or the equipment consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of the parts requiring replacement shall be reported by the contractor to the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

3.7.11 Maintainability. Maintainability Program requirements shall be as described in MIL-STD-470.

3.7.11.1 Preventive maintenance. Preventive maintenance requirements shall be kept at a minimum by using (but not limiting to):

a. Fail-safe features.

b. Components requiring little or no preventive maintenance.

3.7.11.2 Down time. The total mean active down time ,as defined in MIL-STD-721, in the maintenance of the adapter shall be the minimum possible by designing for (but not limiting to):

a. Rapid and positive detection of malfunction.

b. Ease of fault correction.

c. Rapid and positive verification of correction.

3.8 Weight. The adapter shall not weigh more than 7 ounces.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.9.1 Date. The date of manufacture, consisting of quarter and year, shall be legibly stamped with ink in a place where it will not be defaced because of usage. A suitable rubber stamp made in the form of a 1/2 inch square with a horizontal line dividing the square into two segments shall be used for this marking. The upper segment shall contain the number of the quarter, and the lower segment shall contain the last two numbers of the year.

3.10 Workmanship. The adapters shall be manufactured in a thoroughly workman like manner. The finished adapters and all integral parts shall be clean and free of oils and any other materials that might adversely affect the operation of the adapter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions

4.3.1 Gas. The high pressure gas used in testing the adapters shall be oxygen conforming to type I of MIL-O-27210; water pumped nitrogen conforming to type i, class I grade B of BB-N-411; or water pumped air equivalent in dryness of BB-N-411 nitrogen. If nitrogen or air is used, appropriate density corrective factors shall be applied to the flow meter and to the performance of the adapters.

4.3.2 Pressure and temperature. Unless otherwise specified, the test conditions of pressure and temperature will be 29.92 inches of mercury and $77^{\circ} \pm 10^{\circ}$ F, respectively. For any other conditions of pressure and temperature, appropriate corrections shall be made to the test results.

4.3.3 Test set up. Where applicable, the test set up shown in figure 6 shall be used for tests.

4.4 First article inspection. First article inspection shall be performed on four adapters, when first article sample is required (see 3.1 and 6.2). This inspection shall include the examinations and tests as described herein (see 4.6) before regular production is started.

4.4.1 First article. When specified in the contract, two complete first articles shall be submitted to the procuring activity at the time the first article test report is submitted for approval. The Government reserves the right to perform any or all tests, inspections and examinations specified herein.

4.4.1.1 Inspection facility. The first article shall be inspected in a laboratory acceptable to the Government or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity. The required number of first articles shall be stated in the contract. The inspections shall be as described (see 4.6).

4.4.2 Test report. In the event that the first article testing is performed at the contractor's plant, contractor shall, after completion of the first article tests, prepare the first article test report according to MIL-STD-831 (see 6.3).

4.5 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests (see 4.5.1).

b. Sampling plans and tests (see 4.5.2).

4.5.1 Individual tests. Each adapter shall be subjected to the following tests as described (see 4.6):

a. Examination of product.

b. Pressure reduction characteristics (at ground level only).

c. Overpressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. One adapter shall be selected at random from each lot of 100 adapters (or fraction thereof) and shall be subjected to the sampling plan A Tests listed in Table II.

4.5.2.2 Sampling plan B. Two adapters shall be selected at random from each lot of 500 adapters (or fraction thereof) and shall be subjected to the sampling plan B tests listed in Table II. After completion of these tests, the adapters shall pass the tests specified (see 4.5.1).

4.5.2.3 Sampling plan C. Four adapters shall be selected at random from each lot of 1,000 adapters (or fraction thereof) and shall be subjected to the sampling plan C test of Table II.

4.5.2.4 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined.. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.5 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.5.4 Vibration and noises. Vibration, flutter or chatter occurring during breathing or simulated breathing tests shall be cause for rejection of the adapter. If this occurs on sampling test representing a lot of adapters, the entire lot shall be rejected. If the adapter vibrates during continuous flow testing, but the vibration can be stopped by altering the flow or inlet pressures and returning to the test point in question, the adapter shall be acceptable providing the entire test is completed within the limitations prescribed for that test. If the vibration persists after three attempts at a single test point, the adapter shall not be acceptable.

4.6 Inspection methods.

4.6.1 Examination of product. Unless otherwise specified by the procuring activity, each adapter shall be examined for conformance to this specification with respect to workmanship, weight, dimensions and marking.

4.6.2 Pressure reduction characteristics. The adapter shall be connected to a pressure source capable of producing the inlet pressures specified in Table III. The adapter outlet shall be connected to a vacuum source. When the inlet pressures are established in accordance with Table III, the adapter outlet shall be introduced to the vacuum supply in such a manner that the proper flow rates are established. Altitude requirements shall be simulated by placing the adapter in an appropriate environment. When respective inlet pressure (minimum and maximum) flow rates and altitude conditions are established, the outlet pressures, when measured as shown in figure 6, shall fall within the range of outlet pressures shown in Table III for all altitudes.

4.6.3 Overpressure. With the altitude compensating ports plugged and the outlet closed, a pressure of 50 inches of water shall be applied to the inlet; any outward leakage of the adapter shall be cause for rejection. With the altitude compensating ports unplugged and adapter outlet closed, a pressure of inches of water shall be applied to the inlet; outward leakage of the adapter exceeding 100 cubic centimeters per minute shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4.6.4 Low temperature operation. With a test setup similar to the one specified in figure 6, the adapter shall be subjected to an inlet pressure of 3.5 ±0.05 inches of water and shall be stabilized at a temperature of -65° ±2° F. While under these conditions, the adapter shall be subjected to cyclic operation by changing outlet pressures so that flow rates will vary from 0 to 20 to 0 liters per minute. The cyclic operation shall consist of 15 ±5 cycles per minute, each cycle given approximately equal time, and continuing for 50 cycles. Following the cyclic operation, and while the temperature is still stabilized at -65° ±2° F, the adapter shall be subjected to the test specified (see 4.6.2). A 12 percent increase in outlet pressure is allowable for inlet pressures above 25 inches of water.

4.6.5 High temperature operation. Except that the operating temperature shall be stabilized at 160° ±2° F, this test shall be the same as the test specified (see 4.6.4).

4.6.6 Low temperature exposure. The adapter shall be subjected continuously for 48 hours to temperature of -85° ±5° F. This exposure shall be at nonoperating conditions such as ambient inlet and outlet pressures and zero flow rates. After the 48 hour period of exposure, the adapter shall be stabilized to room temperatures and subjected to the tests specified (see 4.5.1).

4.6.7 High temperature exposure. The adapter shall be subjected continuously for 7 hours to a temperature not less than 160° F. This exposure shall be at nonoperating conditions such as ambient inlet and outlet pressures and zero flow rates. After the 7 hour period of exposure, the adapter shall be stabilized to room temperature and subjected to the tests specified (see 4.5.1).

4.6.8 Altitude cycling. At room temperature, zero flow rate, and ambient inlet and outlet pressures, the adapter shall be cycled 100 times in an altitude chamber or equivalent. Each cycle shall start at ground level, go up to 50,000 feet, and return to ground level. This test shall not be completed sooner than 30 minutes with each cycle given approximately equal time. Upon completion of this test, the adapter shall be subjected to the tests specified (see 4.5.1).

4.6.9 G forces.

4.6.9.1 Operating conditions. At ground level and at sufficient inlet pressures to cause flow rates for actual breathing exercises, the adapter shall be subjected to a force of up to 6g while actual or simulated breathing cycles are being conducted. After a sufficient number of breathing cycles, detection of any severe resistance to breathing shall be cause for rejection. Since the basis for acceptance or rejection under this test is personal opinion, the test shall show more than one opinion of the same nature before substantial grounds for rejection of the adapter can be advanced. Upon completion of this test, the adapter shall be subjected to the tests specified (4.5.1).

4.6.9.2 Nonoperating conditions. Under nonoperating conditions the adapter shall be mounted in a suitable device and shall be subjected to a g force of approximately 22g for 0.01 second. The adapter shall be appropriately protected against the g forces that will act if the adapter strikes the ground. Upon completion of this test, the adapter shall be subjected to the tests specified (4.5.1).

4.6.10 Orientation. The adapter shall be subjected to the test specified (see 4.6.2). For this test, the adapter shall be oriented in three positions as follows:

a. Vertical (flow direction up).

b. Vertical (flow direction down).

c. Horizontal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

4.6.11 Reliability. The four adapters (see 4.5.2.3) shall be subjected to a simulated breathing test. The test shall be conducted at room temperature at a rate of between 10 and 20 breathing cycles per minute. Cyclic breathing of the adapters shall be accomplished with peak flow rates of 30 ±5 liters per minute for 90 percent of the cycles and 70 ±10 liters per minute for 10 percent of the cycles (inhalation). During 20 percent of the time, the adapters are subjected to breathing cycles, they shall be vibrated at a frequency ranging from 500 to 2,500 cycles per minute at a double amplitude of not less than 0.018 inch nor more than 0.020 inch. The rate of change of frequency shall be logarithmic and such that a maximum of 15 minutes is required to proceed from 500 cycles per minute to 2,500 cycles per minute back to 500 cycles per minute. When there is no provision for logarithmic cycling, a linear rate of change may be used. The adapter shall be vibrated one-third of the vibration time on the longitudinal axis, one-third of the vibration time on the horizontal axis, and one-third of the vibration time on the lateral axis. The adapter shall be checked at intervals not to exceed 24 hours during reliability testing to determine if a failure has occurred. A failure shall be identified as the inability of the adapter to meet the ground level test point for Pressure Reduction Characteristics (4.6.2). No parts shall be replaced as preventive maintenance during reliability testing; however, failed parts may be replaced. Acceptance will be judged on the basis of (see 4.6.11.1. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.6.11.1 Accept-reject criteria. Acceptance will be judged on the basis of the following:

Failures Permitted	Test Time- Hours	Ratio
1	3,383	3.889
2	4,630	2.661
3	5,812	2.227
4	6,954	1.998

4.6.12 Maintainability verification. A maintainability verification demonstration method as described in MIL-STD-471 shall be selected and submitted for the approval of the procuring activity.

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

* 5.1 Preservation. Preservation requirements for the desired levels of protection will be as specified by the acquisition activity in their procurement document (see 6.2). If requirements are not specified, preserve in accordance with MIL-STD-2073-1.

* 5.2 Packing. Packing requirements for the desired levels of protection will be as specified by the acquisition activity in their procurement document (see 6.2). If requirements are not specified, pack in accordance with MIL-STD-2073-1.

* 5.3 Marking. Marking requirements will be as specified by the acquisition activity in their procurement document (see 6.2). If requirements are not specified, mark in accordance with MIL-STD-129.

5.3.1 Precautionary marking. In addition to the requirements of MIL-STD-129 the following precautionary marking shall appear on each unit container:

"CAUTION: DO NOT ALLOW OIL OF ANY KIND TO BE USED ON OR ABOUT OXYGEN ADAPTERS."

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adapter covered by this specification is intended for use in breathing oxygen pressurizing systems installed in aircraft for adapting high altitude seat kit oxygen to a standard pressure breathing oxygen mask.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. When first article is required (see 3.1, 4.4.1 and 6.3)

d. Levels of preservation, packaging, and packing required (see section 5).

* 6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to tested as specified in 4.4.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results (furnish 2 copies), and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Altitude cycling
G force
Orientation
Overpressure

6.5 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

TABLE I. Pressure reduction characteristics

Flow Range of oxygen (Liters per minute)	Adapter Inlet Pressure (Inches of Water)		Adapter outlet Pressure Range (Inches of Water)	Altitude Range Per 1,000 Feet
	Minimum	Maximum		
0	3.0	4.5	-1.0 thru 3.0	0 thru 29
1 thru 10	2.6	3.6	-1.35 thru 2.0	0 thru 29
11 thru 30	2.0	4.5	-1.4 thru 2.5	0 thru 29
31 thru 50	2.4	4.9	-1.5 thru 2.9	0 thru 29
51 thru 85	3.2	5.7	-1.75 thru 3.5	0 thru 29
86 thru 90	4.2	6.7	-2.0 thru 4.0	10 thru 29
0	3.0	4.5	0.5 thru 4.0	30 thru 37
0	3.0	4.5	0.8 thru 4.0	38 thru 40
0	4.5	11.5	2.0 thru 7.7	41
0	7.7	15.2	3.6 thru 9.1	42
0	10.9	17.9	5.2 thru 10.7	43
0	21.9	29.0	10.7 thru 15.8	47
1 thru 10	2.5	3.5	0.3 thru 2.8	30 thru 39
1 thru 10	2.5	3.6	0.3 thru 3.2	40
1 thru 10	4.5	8.5	2.0 thru 7.2	41
1 thru 10	6.9	11.7	3.2 thru 8.6	42
1 thru 10	10.9	14.8	5.2 thru 10.2	43
1 thru 10	22.7	25.8	11.1 thru 15.3	47
11 thru 30	2.0	4.5	0.6 thru 2.9	30 thru 39
11 thru 30	2.0	4.5	0.6 thru 4.4	40
11 thru 30	4.8	11.5	2.1 thru 7.1	41
11 thru 30	7.8	15.0	3.5 thru 8.5	42
11 thru 30	11.4	17.9	5.4 thru 10.1	43
11 thru 30	23.5	30.0	11.3 thru 15.2	47
31 thru 50	2.6	4.9	0.6 thru 3.0	30 thru 39
31 thru 50	2.4	4.9	0.9 thru 4.5	40
31 thru 50	4.7	11.9	2.0 thru 7.2	41
31 thru 50	7.5	15.6	3.4 thru 8.6	42
31 thru 50	11.3	18.3	5.3 thru 10.2	43
31 thru 50	23.5	29.4	11.2 thru 15.3	47
51 thru 70	4.2	5.7	1.0 thru 3.3	30 thru 39
51 thru 70	4.8	5.7	1.3 thru 4.8	40
51 thru 70	5.5	12.7	2.2 thru 7.4	41
51 thru 70	8.2	16.3	3.6 thru 8.8	42
51 thru 70	12.0	19.0	5.5 thru 10.4	43
51 thru 70	24.0	30.1	11.4 thru 15.5	47
71 thru 90	4.2	6.7	0.9 thru 3.6	30 thru 39
71 thru 90	5.0	6.7	1.2 thru 4.9	40
71 thru 90	4.0	13.5	1.8 thru 7.3	41
71 thru 90	7.2	17.2	3.2 thru 8.7	42
71 thru 90	11.5	19.9	5.15 thru 10.3	43
71 thru 90	24.5	31.0	11.0 thru 15.4	47

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

TABLE II. Sampling plan

Inspection	Method
Sampling Plan A 4.5.2.1	
Low temperature operation	4.6.4
High temperature operation	4.6.5
Pressure reduction characteristics (at all altitudes)	4.6.2
Sampling Plan B 4.5.2.2	
Low temperature exposure	4.6.6
High temperature exposure	4.6.7
Sampling Plan C 4.5.2.3	
Reliability	4.6.11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

TABLE III. Pressure reduction characteristics test point

Altitude Per 1,000 Feet	Flow of Oxygen (Liters Per Minute)	Adapter Inlet Pressure (Inches of Water) Minimum	Adapter Inlet Pressure (Inches of Water) Maximum	Adapter Outlet Pressure (Inches of Water)
ground level	0 1/	3.0	4.5	-1.0 thru 3.2
	10	2.6	3.6	-1.35 thru 2.0
	50	2.4	4.9	-1.5 thru 2.9
	85	3.2	5.7	1.75 thru 3.5
30	0 1/	3.0	4.5	0 5 thru 4.5
	10	2.5	3.5	0.3 thru 2.8
	50	2.6	4.9	0.6 thru 3.0
	90	4.2	6.7	0.9 thru 3.6
41	0 1/	4.5	11.5	2.0 thru 7.7
	10	4.5	8.5	2.0 thru 7.2
	50	4.7	11.9	1.9 thru 7.2
	90	4.0	13.5	1.4 thru 7.3
47	0 1/	21.9	29.0	10.7 thru 15.9
	10	22.7	25.8	11.1 thru 15.3
	50	23.5	29.4	11.2 thru 15.3
	90	24.5	31.0	11.0 thru 15.7

1/ The outlet pressure for zero flow rates shall be maintained for one minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

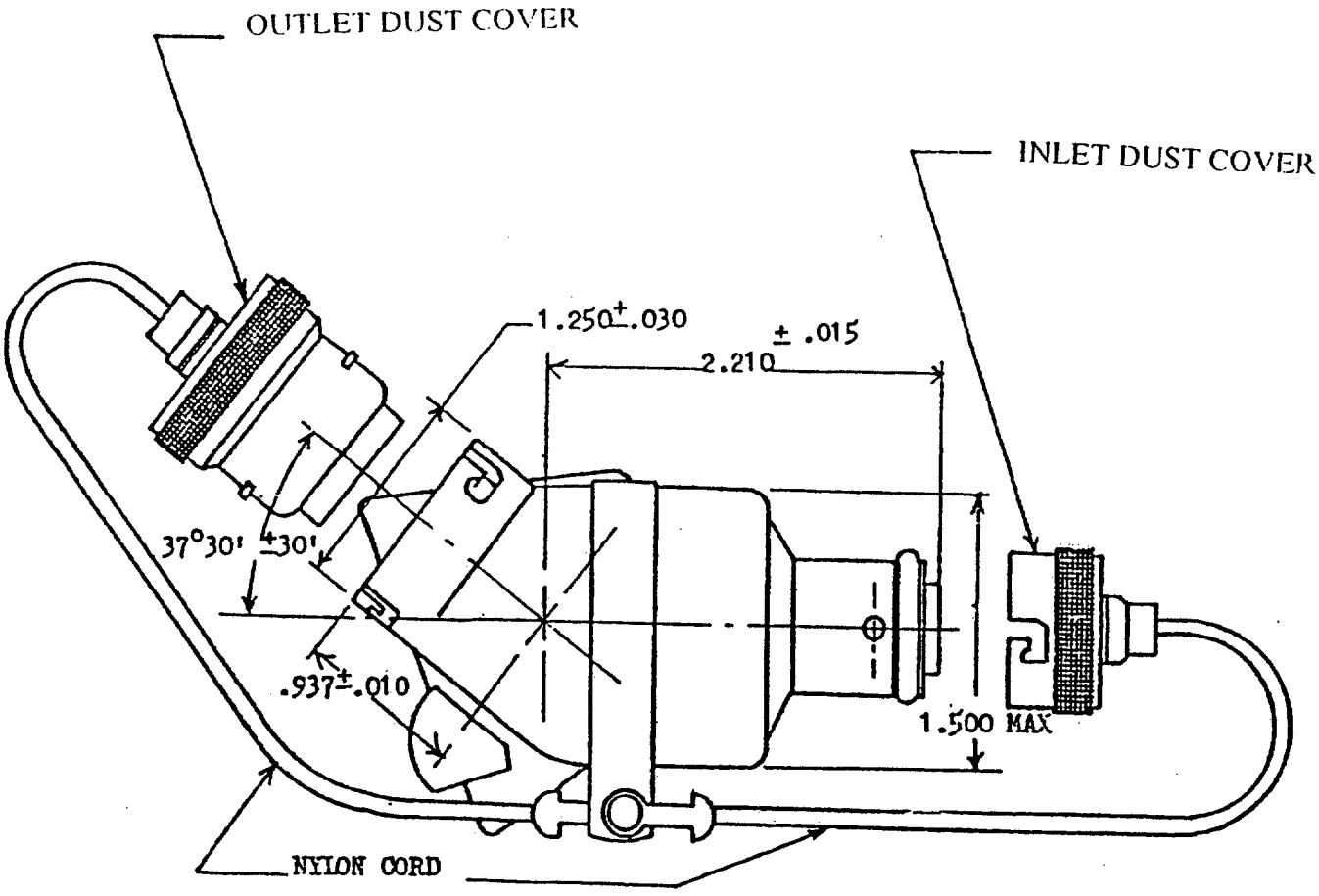
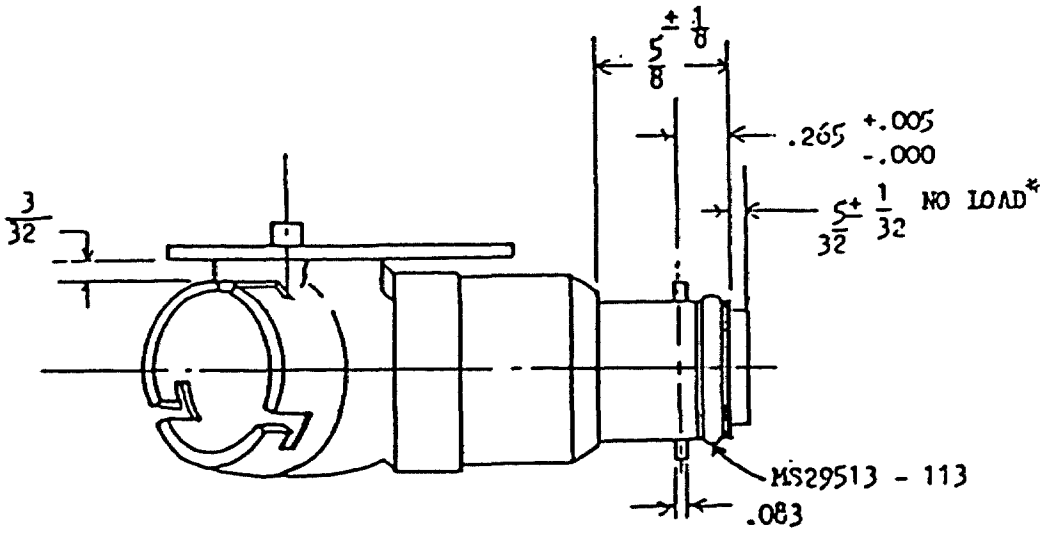
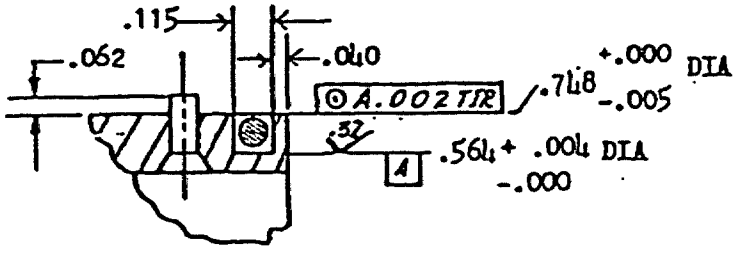


FIGURE 1. Adapter

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.





ENLARGED SECTION AT MALE END

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE INCHES AND TOLERANCES ARE:

FRACTIONS	DECIMALS	ANGLES
± 1/64	±.005	± 1/2°

* LOAD TO DEPRESS SLEEVE FLUSH WITH END OF PORT SHALL BE 5 +1 POUND

ALL SURFACES 63 FINISH PER USAS 46.1 UNLESS OTHERWISE NOTED.

FIGURE 2. Adapter inlet

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

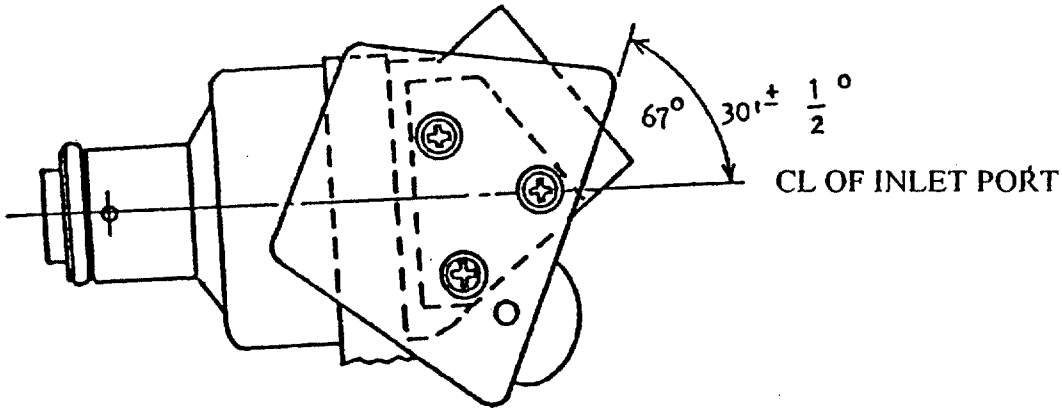


FIGURE 3. Plate alignment

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

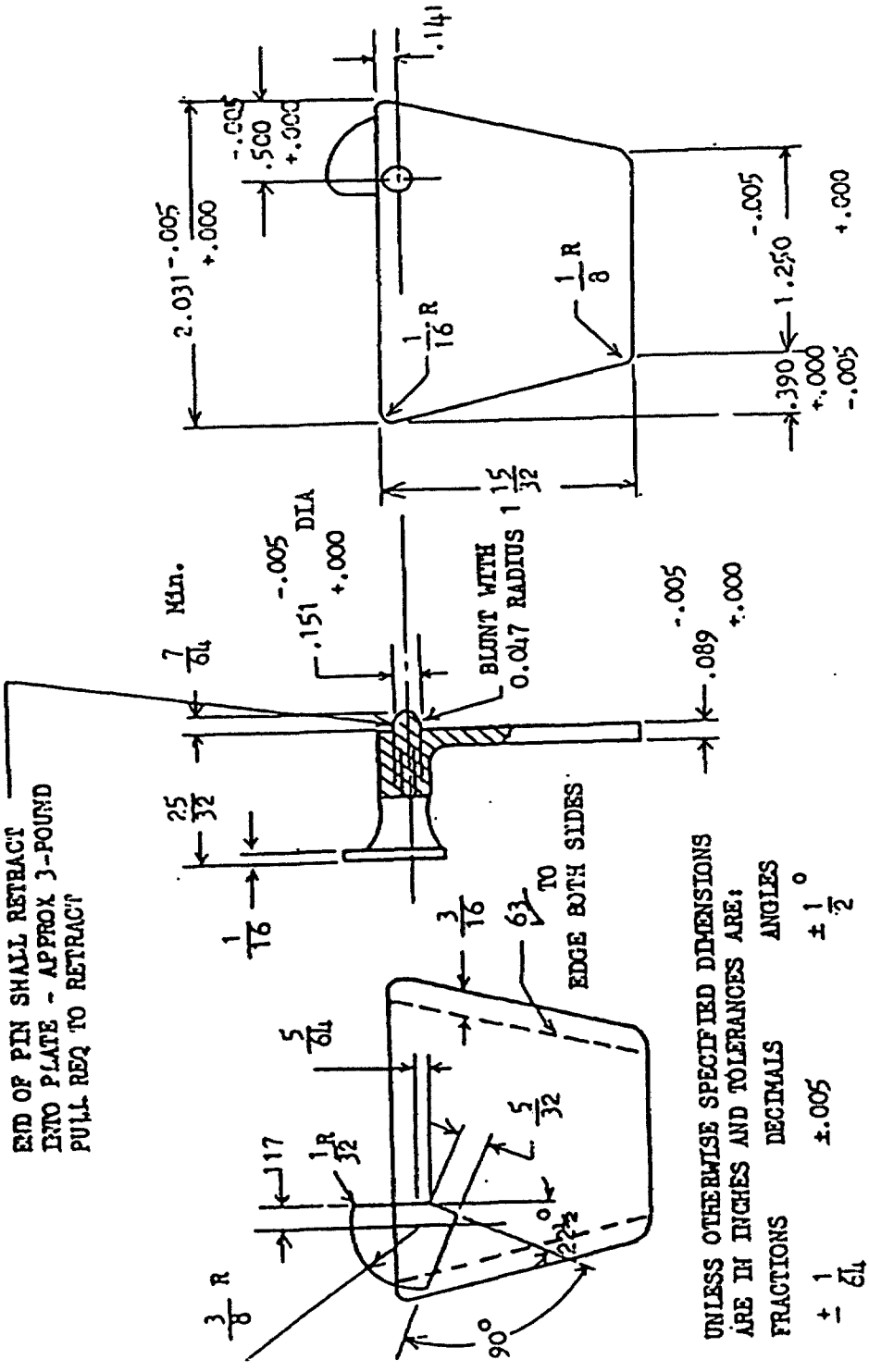


FIGURE 4. Plate and locking pin assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

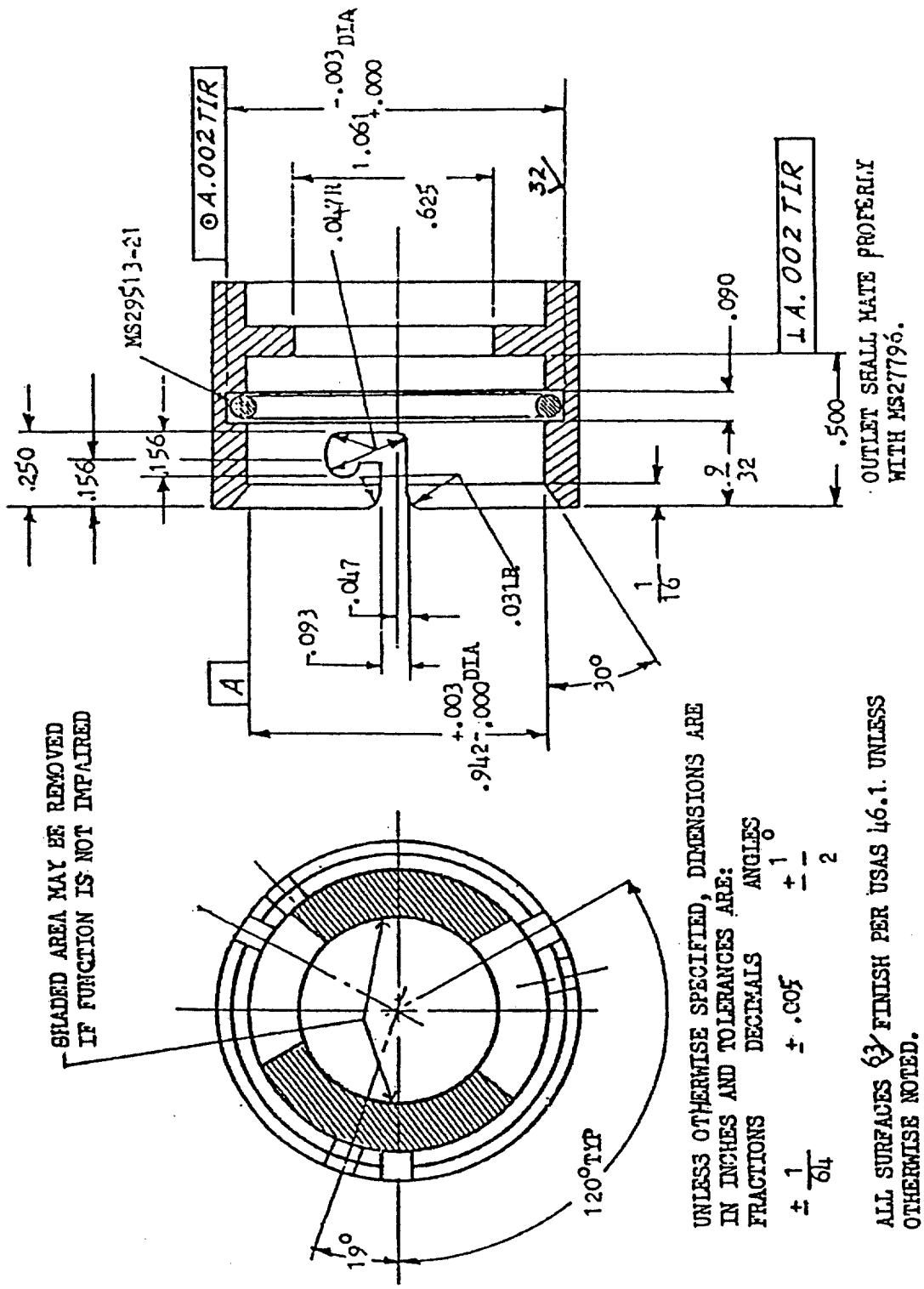
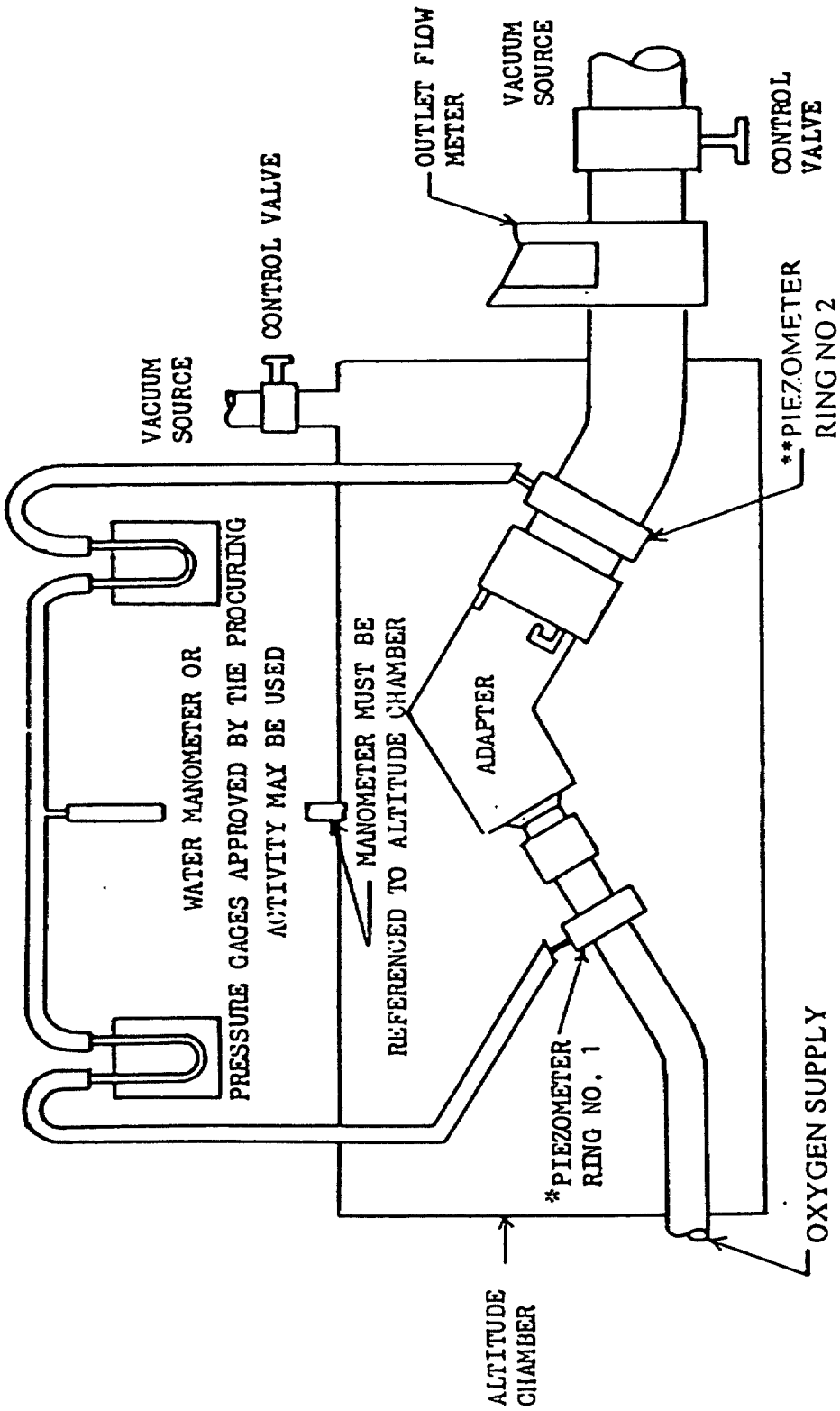


FIGURE 5. Adapter outlet

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.



* PIEZOMETER RING NO. 1 MAY BE CONNECTED TO AIRLOCK, INC. FITTING NO. 265
** PIEZOMETER RING NO. 2 MAY BE CONNECTED TO A FITTING IN CONFORMANCE TO 5783661

FIGURE 6. Test setup for pressure reduction characteristics

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.

Custodian:
Air Force - 99

Review activity:
Air Force - 11

Preparing activity:
Air Force - 82

Agent activity:
Air Force - 99

(Project 1660-F643)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:06Z
Check the source to verify that this is the current version before use.